

IVL Workflow - A Blueprint to Designing Applications for an Immersive Visualization Laboratory

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ABSTRACT

Immersive Visualization Laboratories (IVLs) have existed for over 50 years, highlighting the longevity of these laboratories. While there are some reviews and reports on long-term IVLs, there is little research on the workflows within such a center. One major factor that contributes to maintaining an IVL are workflows. While there are many ways to create immersive visualizations (IVs), few laboratories report on their workflows for creating their IVs. Based on prior research and experience maintaining an IVL and different workflows, we present our current workflow for an IVL. Our goal is to report our process, so that it may aid a broad audience, people who are interested in creating an IVL, and start a conversation between IVL experts.

Index Terms: Virtual Reality, Large-Scale Immersive Installation, Immersion, Workflows

1 INTRODUCTION

Immersive Visualization Laboratories (IVLs) have existed for over 50 years [12]. These laboratories encompass (generally) a physical space where one or a team of visualization and Virtual Reality (VR) experts develop and maintain software and hardware corresponding to immersive visualizations (IVs) [18, 22, 15]. There are some reports from IVLs on their experiences over the years, which highlight the individuality of the laboratories. This paper however aims to identify a standard or basic workflow of an IVL.

Our aim is to give a broad audience an overview of the workflow in an IVL and to initiate discussions between IVLs. We also intent to showcase the effort and process to a broad audience and policymakers. While each laboratory has its own specialty, a basic workflow for a standard use case may help demystify the work that goes into an IVL.

The paper is structured as follows: First, we look at IVs then highlight some IVLs and their workflows. Based on this, we propose our concept for an IVL workflow (that is applicable to IVLs and IVs in general), then we showcase the workflow with a game engine and apply the workflow to an active use case.

Finally, we discuss the limitations of the IVL workflow, our findings on the workflow, and actionable steps.

2 BACKGROUND

Scientific visualization (SciVis) or the visualization of scientific data can benefit from VR through the interactive and spatial exploration of complex data [1].

Visualizations can also support domain scientists in their research but additionally communicate their findings to the public through science communication [11].

IVs can assist in SciVis [1, 16] but also in the learning effect when educating a broader audience: “Learners experience higher presence, absorption, flow, psychological immersion and positive emotions in a virtual reality setting compared to a desktop setting” [26]. One way to create and experience IVs is through an IVL.

2.1 Creating Immersive Visualizations

Many scientific fields can benefit from IVs [16], however, there are many solutions when it comes to creating an IV [17], e.g. software specifically for SciVis [4] or game engines [18]. One vital aspect for creating IVs is the real-time and interactive factor [1, 17]. The interactions should be specific to the visualization, either for exploration [1] or annotation [17].

One example of an IV workflow would be from Zhang et al. [33]. Zhang et al. summarized the process of creating IVs into a few steps [33]:

1. Raw Data
2. Dimension Reduction, Aggregations
3. Processed Data
4. Feature Encoding
5. Visualization Presentation
6. Interactive Feedback
7. User view

Another example would be from Tucker et al. [30], where the visualization process is broken down into seven steps: Gathered data, data filter, representation mapping and rendering, which is influenced by viewport, display and lighting.

There are some reports of case specific workflows [31], where a workflow to bring computational fluid dynamics into immersive environments with the help of a game engine was proposed. Korkut et al. found “With the myriad of advantages provided for visualization and virtual reality, most of the studies prefer to use game engines.” [16].

On the software side, IVs can be created from game engines [18, 14] to custom software [12] to immersive SciVis software like immersive ParaView [29] or COVISE [4].

These workflows indicate, while there is a variety in software, there is a way to break down the workflows to be applicable to many use cases.

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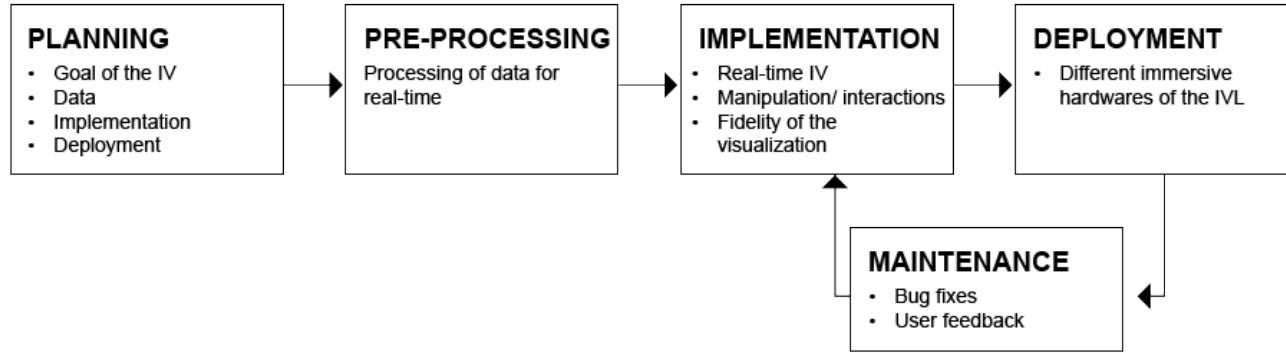


Figure 1: Overview of the IVL Workflow: Planning, Pre-Processing, Implementation, Deployment, and Maintenance

2.2 Immersive Visualization Laboratories

IVLs are physical laboratories, where one or a team of visualization and VR experts develop and maintain software and hardware corresponding to IVs [18, 22, 15, 12]. However, each IVL is very unique, for example in its hardware, software, team composition and their workflows.

Hardware for IVLs can range from different output devices like large-scale immersive installations [12, 18, 22, 19] or CAVEs [3, 14] to head-mounted displays (HMDs), depending on their respective focus [22, 14]. Large-scale immersive installations can range from 2-5 walls [24] and are unique to the IVL where they are installed. These installations can offer co-located collaboration but have challenges, e.g., cost and maintenance [24, 4].

However, IVLs can feature HMDs or other hardware to offset these challenges when providing IVs [18, 14]. “CAVE VR system has numerous advantages, but because of the difficulties in running software on the system, scientists and researchers could be unwilling to utilize it. These issues could have to do with the system’s incompatibility with particular programs or its difficulty being properly set up. Therefore, researchers and scientists may think about utilizing different VR systems that offer a more streamlined software experience.” [14]. IVLs can also have a wide array of input devices like omni treadmills [14].

Some IVLs have published reviews reflecting on their learnings ranging from 10 years [22], 15 years [15], 25 years [21], and 50 years [12]. These reviews highlight the differences in hardware but also cover other aspects, from general learnings to software infrastructure and some workflows.

2.3 Workflows within IVLs

The IVL reports encompass many features of the laboratories, amongst them are software infrastructure [14, 18, 22, 12], but only a few discuss their workflows. Krüger et al. took an in depth look at using game engines, specifically Unreal Engine (UE) for their IVL work [18]. A few IVLs use game engines for their process. These game engines can be approachable specifically for IVs their application depends on the use case [18, 6].

3 IVL WORKFLOW

In a 50 year anniversary paper of an IVL [12], Johnson et al. reflect on maintaining a laboratory for so long. Their lessons learned include flexibility with workflows and tools. The necessity to keep tools up-to-date and regularly work out bugs. Their experience also shows the need to embrace evolution (software and hardware).

In another report, Khadka et al. highlight their workflow that is based on a SCRUM workflow [15]: setting project goals, assembling a team, organizing the SCRUM process.

There is little research or reports on a basic workflow for an IVL. While there are some reports on workflows, these are either software specific [18] or IVL specific [22, 12]. This paper proposes an IVL workflow / a blueprint for a basic workflow for creating applications in an IVL. For this we first propose our concept then apply the IVL workflow to a game engine and highlight the workflow through a use case.

Our report builds on application design based on user-centered design [7], SciVis [30], IVs [33] and our own [redacted to keep anonymity] years of experience maintaining an IVL. Our workflow is based on applying game development elements including the game design document (GDD) to creating applications [32, 25]. The GDD is a document that accompanies a game development and encompasses all aspects of development from goals to implementation. The GDD has been applied to general software development [27] and scientific visualization before [32].

Over the course of 12 years, we have used various approaches: custom software solutions, SciVis software, and game engines [22]. Similar to Krüger et al. [18] we discovered that game engines offer the toolset necessary for creating IVs for IVLs. Game engines provide good rendering quality, quicker development time and have a lower entry barrier for the team or students. While the IVL workflow can be applied to any software, we will be using a game engine.

3.1 Concept

The workflow for application development for IVLs is similar to that for immersive applications [2, 28], SciVis applications [30], user-centered design [7] and game development [25].

There are many other resources that dive into a detailed creation of an immersive application [2, 28, 33] and the creation of SciVis [30]. We encourage readers explore those resources for details suitable to the visualization they are creating.

Our IVL workflow is for a broader audience and is broken down into a few steps, similar to the basic visualization workflow [30] and IV workflow [33].

When comparing IV workflows from different fields [17, 31, 33, 14] to workflows within immersive laboratories [18, 22] we propose the following workflow (see Figure 1).

Planning: The first step is a discussion between the IVL and the domain scientists. This step can be a time consuming task but from our experience it is vital for the creation and design of an IV within an IVL. The IVL team or expert have their experience with IVs and should communicate the possibilities and potential challenges with the domain scientists. This way all involved can gain insight and give feedback. Here, we recommend creating a GDD [25], a technical document or similar document that encompasses all decisions made and goals defined.

Within this document the following topics need to be discussed (see Figure 1 PLANNING): (1) The goal of the overall IV, (2) data, (3) implementation and (4) deployment. Interactions and functionality [33] and the fidelity [20] of the IV should be discussed and defined within the planning step and documented within the (3) implementation chapter of the technical document. The target output for the IV should be defined within the technical document during planning as it influences both the (3) implementation and the (4) deployment.

Once the planning is completed, the technical document has all information necessary for the next step. This document should also accompany the entire development and be updated accordingly.

Pre-Processing: The next step is the pre-processing of the data (see Figure 1) to display it in a real-time immersive setting [1, 18]. For SciVis this is also the data filtering step [30]. This is either in the form of data reduction if the data is too large for real-time display or filtering if the data requires it [5, 14]. Pre-processing is unique to each IV, dataset and scientific domain. When using a game engine in the workflow one major bottleneck lays within the pre-processing of data [22]. Here, it is vital to know the capabilities and feature set of the software and and prepare the data accordingly for the implementation.

Implementation: Implementation is the main step of creating an IV (see Figure 1 IMPLEMENTATION). Based on prior research, this step focuses on the real-time visualization of the data as well as the manipulation or interactions with the data. The implementation also focuses on the fidelity of the visualization [20]. Different IVL software offer a wide range of interaction tools for immersive applications for basic interactions like navigation, it may be necessary to implement custom functions depending on the IV.

Deployment: IVLs have the special use case of having the possibility to deploy IVs on various immersive systems [17]. Some of the workflows report on the importance of discussing the portability of the application [17]. Deployment options can be individual for each IVL depending on their hardware and for each project based on the requirements from the technical document. The target output hardware range from one HMD to a large-scale immersive display (see Figure 1).

While many IV software have the capability of deploying a project onto various devices with different operating systems, the work lies within translating the interactions for the application to each device as well as adapting the application to run with good performance on the target device. As IVLs can have a multitude of hardware for applications, the visualization should be deployable on different systems [18] that were defined in the planning step.

Maintenance: The final step for IVLs would be the maintenance of the IV. This could include bug fixes or adding new features [12] to updated datasets to changes based on user feedback [7]. The changes from maintenance then feed back to the Implementation step, which then requires a new deployment of the IV (see Figure 1).

3.2 Use Case: Real-Time Lighting Simulation in a Historic Room

This section highlights the application of the blueprint for an active use case. The use case presented is in a collaboration with the University of Bamberg, where the castle hall in Rheinsberg, Germany, was captured using photogrammetry [8, 10, 9], and an interactive immersive experience was needed to accurately recreate the mirrors and lighting in the room. For this project, we use UE 5.4¹.

Planning: The first step was the formalization goals and requirements of the IV:

1. Goal: A photo-realistic representation of the castle hall with its mirrors and the ability to switch between different lighting

¹Unreal Engine Homepage



Figure 2: Lighting Scenario: Day Time



Figure 3: Lighting Scenario: Night Time

scenarios in real-time to be used by art historians to analyze the frescoes and paintings in the room but also for science communication of their findings.

2. Data: Photogrammetry model
3. Implementation: Next to a basic navigation interaction there was the need to switch between different lighting scenarios.
4. Deployment: The target output hardware encompass a 5-sided CAVE [23], tethered VR HMD and mobile VR HMD.

Next the fidelity of the application, where a photorealistic visualization was required. There were no use case-specific interactions.

Pre-Processing: The next step was pre-processing the raw photogrammetry data. Pre-processing for the castle hall was performed in Blender², where the floor was reworked to remove artifacts from the photogrammetry scan, and the mirrors were replaced with reflections. Then materials for mirrors, floor and windows were reworked to suit a physically based rendering workflow [13]. This way material properties could be transferred from Blender to UE.

Implementation: Once the castle hall was in UE, it was a matter of implementing the different lighting scenarios for the IV. The scenarios are day (see Figure 2) and night (see Figure 3). One main interaction of the IV was for users to be able to navigate through the castle hall. For this a flying and teleport navigation method were implemented. Another interaction required was a function, where users could switch between different light simulations.

Deployment: The castle hall runs on a five-sided CAVE system based on LEDs [23], tethered VR and mobile VR. Depending on the output device the interactions are mapped for the various inputs (see Table 1).

The differences between the applications are not only the interaction mapping but also the optimization for the different target hardware. With CAVE systems, it is common for them to be steered by

²Blender Homepage

Target Hardware	Input	Computation
CAVE	ART Flystick	Linux Cluster
Tethered VR	HMD controller	Windows PC
Mobile VR	HMD controller	Meta Quest 3

Table 1: Overview of the IVL Workflow: Deployment of the Castle Hall on various Hardware

a computer cluster [23], where it is essential to be able to distribute the IV on multiple devices and maintain a good performance. Comparably a tethered VR system is either a desktop PC or a laptop. These two target output devices have more power compared to a mobile VR version. The mobile VR version has to be optimized for a smaller computing power.

Maintenance steps include feedback from the art historians as well as the feedback from users on the lighting scenarios.

4 LIMITATIONS AND DISCUSSIONS

A major limitation of this paper is that it is based on our experience in the IVL and only highlights one use case. The use case we chose is very straight forward and doesn't require custom software. However, we wanted to showcase the use of the IVL workflow with a basic use case that IVL might face.

Another limitation of the paper is that the basic workflow ends with the maintenance step. We propose expanding the maintenance step with an evaluation [11]. The evaluation should be formal evaluation of the IV to gain a better understanding on if the IV fulfills the goals and is usable. This step could help refine the visualization and potentially help the IVL with future IVs.

In that spirit, another limitation is a formal evaluation of the IVL workflow and its applicability on different IVL use cases.

We applied a game engine to our use case and have identified the following advantages and challenges of using a game engine for an IVL, these align with what Krueger et al. found [18]. One major advantage to using game engines in an IVL is *approachability*. Game engines nowadays offer documentation and can be used at an entry-level. Krueger et al. report that they required less time to train students and colleagues to use UE [18] which also encompasses their two requirements accessibility and wide adoption.

Another major advantage to the IVL workflow is code *reusability*, depending on the projects of the IVL there is a high chance of projects being similar. The workflow prevents starting from scratch every project. Based on learnings and evaluation of prior projects, new projects could be completed faster. Secondly, the pre-processing of data can help the IVL teams gain knowledge on data processing and speed up the process for new projects.

A major challenge of the IVL workflow is the validation and applicability of the workflow within various IVLs, which we will address in a next step.

5 CONCLUSION AND FUTURE WORK

The goal of this paper is to present a basic workflow for creating an IV at an IVL to a broad audience so that on the one hand people unfamiliar with IVL may gain a better understanding on how these laboratory works but on the other hand that there may be a general discussion on workflows in laboratories from VR expert. In the second group, we hope that anyone who wants to build an IVL can take away a better understanding on creating IV applications but also the work that goes into an IVL.

In the future, we hope that experts from IVLs may share their workflows to help people interested in creating and maintaining one. IVLs have existed for a while and encompass many more aspects, such as financial sustainability, technical sustainability, and staff retention [15]. We conclude that the next steps should include a scoping review of IVLs and what factors could contribute to their longevity.

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